S.B MUTHU KUMAR

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PROFESSIONAL SUMMARY

Detail-oriented Computer Science and Engineering student with strong hands-on proficiency in Python, Java, C/C++, full-stack web architectures, and relational databases. Experienced in developing multi-agent workflows, RESTful backend APIs, and interactive 3D graphics in Unity. Demonstrated competence in software engineering principles, algorithm design, and containerised deployments.

TECHNICAL SKILLS

Programming Languages: Python, Java (JDK 17 LTS), C, C++, SQL, JavaScript

Frameworks & Web: Flask, FastAPI, HTML5, CSS3, Streamlit, RESTful APIs

Databases & Tools: MySQL, SQLite, XAMPP, Git, GitHub, Docker, MSYS2 MinGW-w64

AI & Data Technologies: LangChain, LangGraph, Sentence-Transformers, spaCy, PyPDF

3D & Interactive Media: Unity 6 LTS, C#, Blender, GLB/GLTF assets, Adobe After Effects

KEY PROJECTS

Autonomous Agentic AI System for Resume Tailoring & JD Alignment | *Python, LangGraph, Embeddings*

- Engineered an autonomous multi-agent state machine utilizing LangGraph and Sentence-Transformers to dynamically tailor resumes against specific technical job descriptions.
- Designed an automated deterministic diff-checking evaluator that cross-references drafted claims with master ground truth, enforcing a 0% credential hallucination rate.
- Implemented programmatic PDF compilation using ReportLab, generating standardized ATS-compliant resumes with elevated semantic keyword alignment.

Smart Museum & Cultural Heritage Explorer in AR/VR | *Unity 6 LTS, C#, Blender, URP*

- Developed an immersive 3D cultural exploration application using Unity 6 LTS with Universal Render Pipeline (URP) and AR Foundation.
- Optimized complex 3D heritage artifacts and mesh displacements in Blender, exporting lightweight .glb/.gltf formats for real-time mobile rendering.
- Programmed responsive C# UI interaction triggers and custom camera navigation rigs for virtual museum environments.

Disaster Management Multi-Agent Coordination System | *Python, AI Agents, REST APIs*

- Collaborated in a 24-hour hackathon to build a distributed multi-agent dispatch system coordinating real-time disaster alerts and resource allocation.
- Implemented modular task routing algorithms and structured JSON telemetry communication between autonomous agent nodes, securing 2nd runner-up prize.

EDUCATION

Bachelor of Engineering in Computer Science and Engineering (B.E. CSE)

Sathyabama Institute of Science and Technology, Chennai, India | Expected Graduation: 2028

- Relevant Coursework: Data Structures & Algorithms, Database Management Systems (DBMS), Object-Oriented Programming (Java/C++), Operating Systems, Software Engineering.